

Sushant Mishra

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Aspiring Tech Innovator | Electrical Engineering Student | Hardware & VLSI Enthusiast

Electrical Engineering student at Delhi Technological University with a deep passion for semiconductor engineering, VLSI design, and hardware innovation. Experienced in rapid prototyping with Arduino and FPGA, and excited by the intersection of AI/ML with chip design and intelligent hardware acceleration. Proven ability to bridge the gap between custom hardware RTL architectures and full-stack software integration.

Education

Delhi Technological University (Formerly DCE) <i>B.Tech Electrical Engineering</i>	New Delhi, India <i>Jul 2025 – Jul 2029</i>
Kendriya Vidyalaya <i>Secondary & Senior Secondary Education</i>	India <i>Mar 2012 – Mar 2024</i>

Experience

Houston International Foods Pvt. Ltd. <i>Full-Stack Web Developer Intern</i>	Mumbai, Maharashtra <i>Jun 2025 – Jul 2025</i>
- Developed and deployed web applications for the Information Technology department, streamlining internal data flows. - Executed frontend-backend architecture design, demonstrating strong problem-solving skills and clean code execution. - Recommended by the organization for future opportunities in website development due to punctual delivery and collaborative teamwork.	

Projects

FPGA-Accelerated Edge Intelligence System <i>PYNQ-Z2 (Zynq-7020), ESP32-S3, VHDL, C, UART</i> SemiXthon 2026	
- Engineered a Tiny Edge AI Accelerator capable of real-time anomaly detection and pattern recognition at the edge with zero cloud dependency. - Designed a custom RTL processing pipeline (Threshold Detector, FSM Classifier, Pattern Matcher) that achieved a verified per-frame latency of 57.17 μs, operating 87x faster than the required 5ms target. - Established a high-speed (921600 baud) UART communication pipeline between the ESP32 sensor host and the FPGA processing core.	
Advanced Driver Assistance Systems (ADAS) <i>Python, OpenCV, PyTorch, C++, Sensors</i>	
- Architected an integrated ADAS pipeline focused on real-time data processing and computer vision. - Fused hardware sensor data with machine learning object detection models to evaluate real-time environmental conditions and reduce processing latency.	
Hand-Gesture Recognition via Custom RTL Accelerator <i>FPGA, Verilog/VHDL, Python</i>	
- Designed and implemented a Custom Register-Transfer Level (RTL) architecture specifically optimized for edge AI logic. - Shifted traditional software-based AI recognition to a dedicated hardware accelerator, drastically reducing power consumption and inference time.	

Hackathons & Achievements

AM Hacks 2.0 (AssetMerkle IGDTUW) <i>Top 20 Finalist (Team SJJ)</i>	New Delhi, India <i>Feb 2025</i>
- Ranked in the Top 20 out of all competing teams in a highly competitive 48-hour offline hackathon. - Served as the sole technical lead for the team, orchestrating rapid development and clean execution. - Architected and built the entire prototype web app (exclusively software/frontend UI) during a major strategic pivot, demonstrating high-pressure problem-solving capabilities.	

Technical Skills

Hardware Platforms: FPGA (PYNQ-Z2), ESP32, Arduino Nano/Uno
Semiconductor & VLSI: Chip Design, Logic Gate Architecture, Transistor-level Design, VLSI Design, Verilog, VHDL/RTL
Programming Languages: C++, Python, C, JavaScript
AI/ML: AI-driven Hardware Acceleration, Automated Layout Optimization, TinyML/Edge AI, PyTorch, Scikit-learn, NumPy
Web Development: Full-Stack Development, Frontend-Backend Architecture, HTML/CSS
Core Concepts: Digital Logic Design, Embedded Systems, Rapid Prototyping, Data Structures & Algorithms